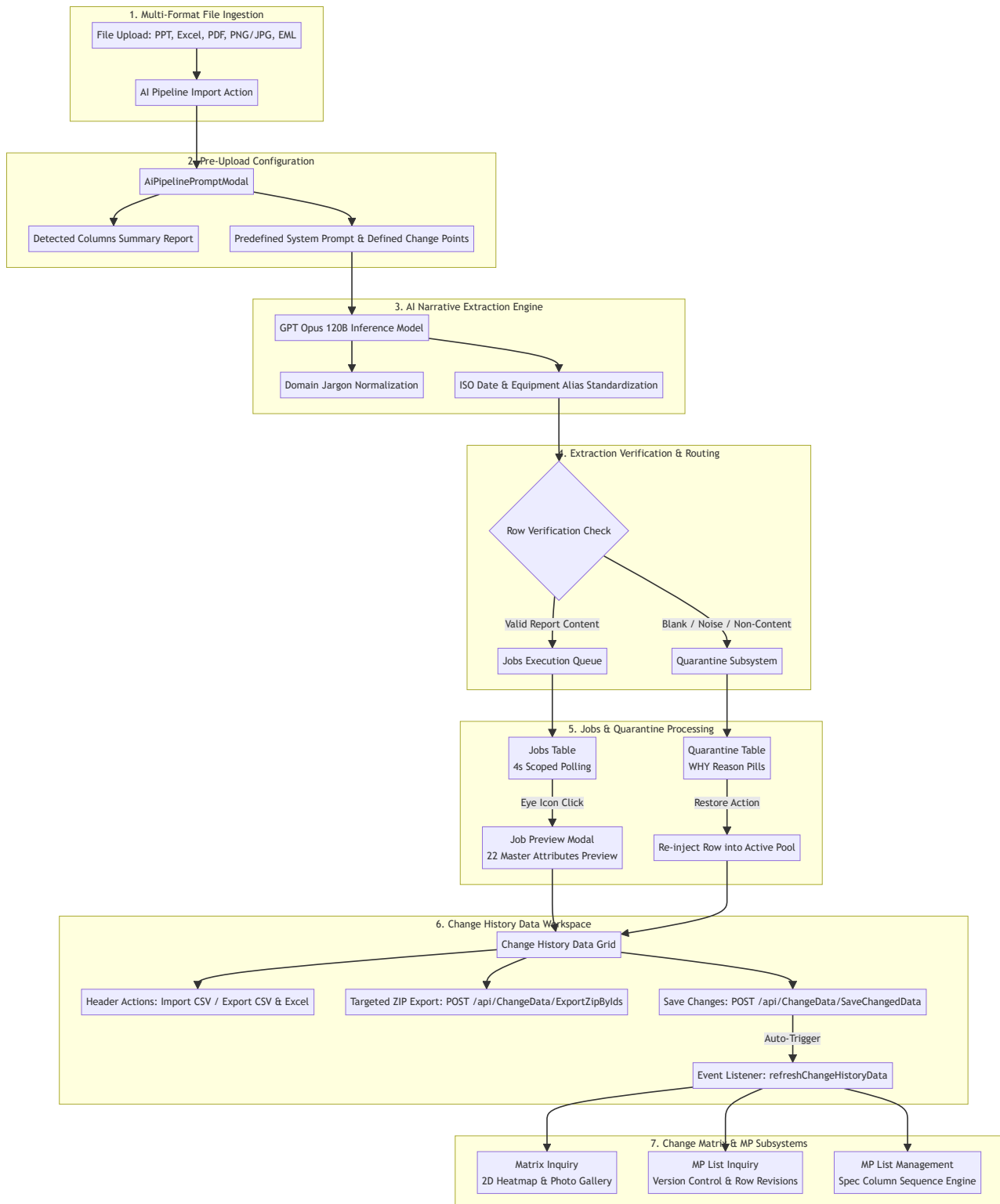


# System Specification: End-to-End Flow Diagram & Data Lifecycle

## 1. System Architecture & Workflow Diagram

This document illustrates the complete end-to-end operational flow of the EQUAL system—from initial multi-format file upload through AI narrative extraction, job monitoring, quarantine handling, master schema standardization, change history management, matrix heatmap visualization, and maintenance plan (MP) versioning.

 **SYSTEM WORKFLOW & PROCESS ARCHITECTURE**



## 2. Stage-by-Stage Operational Summary for QC Testing

| Stage # | Stage Name                     | System Action                                                   | QC Verification Expectation                                    |
|---------|--------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| 1       | <b>Multi-Format Ingestion</b>  | Ingests .xlsx , .csv , .pdf , .pptx , .png , .eml files.        | System accepts all 5 supported file types without rejection.   |
| 2       | <b>Pre-Upload Report</b>       | Displays column headers and editable prompt with Change Points. | Column summary tags match input file headers exactly.          |
| 3       | <b>AI Extraction</b>           | GPT Opus 120B maps text to 22 master schema columns.            | Jargon normalized; dates converted to ISO-8601 ( YYYY-MM-DD ). |
| 4       | <b>Routing &amp; Isolation</b> | Valid rows sent to Jobs; blank/noise sent to Quarantine.        | Invalid/empty rows isolated without breaking the job run.      |
| 5       | <b>Jobs &amp; Quarantine</b>   | 4s polling on Jobs; Restore action on Quarantine.               | Polling active only on Jobs page; Restore returns row to pool. |
| 6       | <b>Change History Data</b>     | 22-attribute grid editing, targeted ZIP export by IDs.          | Selected checkbox IDs exported in ZIP package.                 |
| 7       | <b>Matrix &amp; MP Modules</b> | Heatmap status badges, MP versioning, column sequence.          | Column order strictly matches sequence numbers.                |